

MTH001: Elementary Mathematics - Comprehensive Conceptual Preparation Document

Modules 16 to 30 - Complete Study Guide

Document Information

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Quranic Motivation

"Insaan ke liye wahi kuch hota hai jis ke liye woh khud mehnat aur koshish karta hai." **Translation:** "For a person, only that which they strive and work hard for comes to pass." **Reference:** Surah An-Najm, Ayah 39

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Introduction {#introduction}

This comprehensive preparation document covers all essential concepts from MTH001 modules 16 to 30. Each module is explained using simple language with practical daily life examples to ensure clarity and retention. The document includes important definitions, formulas, and a complete set of 100 quiz questions with detailed explanations.

How to Use This Document:

1. Read the module explanations carefully and understand the concepts
 2. Review the daily life examples to connect theory with practice
 3. Study the key definitions and formulas
 4. Attempt the quiz questions to test your understanding
 5. Review the explanations for any questions you find difficult
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Module Explanations {#module-explanations}

Module 16: Percentage Change

Author: ZB ZAIB (Module 16)

Percentage change measures how much a value has increased or decreased relative to its original value. This concept is fundamental in finance, commerce, and everyday calculations.

Daily Life Example 1: Mobile Phone Price

Imagine you purchased a mobile phone for Rs. 20,000 last year. This year, the same model costs Rs. 24,000. To find the percentage increase:

- $\text{Change} = 24,000 - 20,000 = \text{Rs. } 4,000$
- $\text{Percentage Change} = (4,000 / 20,000) \times 100 = 20\%$

This means the price has increased by 20% compared to last year.

Daily Life Example 2: Salary Increase

Your monthly salary was Rs. 50,000. After a promotion, it becomes Rs. 60,000. The percentage increase is:

- $\text{Change} = 60,000 - 50,000 = \text{Rs. } 10,000$
- $\text{Percentage Change} = (10,000 / 50,000) \times 100 = 20\%$

Your salary has increased by 20%.

Daily Life Example 3: Weight Loss

You weighed 80 kg and after exercise and diet, you now weigh 72 kg. The percentage decrease is:

- $\text{Change} = 72 - 80 = -8 \text{ kg}$
- $\text{Percentage Change} = (-8 / 80) \times 100 = -10\%$

You have lost 10% of your body weight.

Module 17: Interest and Discounts

Author: ZB ZAIB (Module 17)

This module covers two important financial concepts: interest (money earned on investments or paid on loans) and discounts (reductions in price).

Simple Interest

Simple interest is calculated only on the original amount (principal), not on any interest already earned.

Formula:

Daily Life Example: Bank Savings

You deposit Rs. 10,000 in a savings account at 5% simple interest per year for 3 years. The interest earned is:

- Simple Interest = $10,000 \times 0.05 \times 3 = \text{Rs. } 1,500$
- Total Amount = $10,000 + 1,500 = \text{Rs. } 11,500$

You earn Rs. 1,500 in interest over 3 years.

Compound Interest

Compound interest is calculated on both the principal and previously earned interest, resulting in faster growth.

Formula:

Daily Life Example: Long-Term Investment

You invest Rs. 10,000 at 5% compound interest per year for 3 years:

- Year 1: $10,000 \times 1.05 = \text{Rs. } 10,500$
- Year 2: $10,500 \times 1.05 = \text{Rs. } 11,025$
- Year 3: $11,025 \times 1.05 = \text{Rs. } 11,576.25$

Discounts

A discount is a reduction in the original price, usually expressed as a percentage.

Form

Daily Life Example: Shopping Sale

A shirt originally costs Rs. 1,000 with a 20% discount. The selling price is:

- Discount = $1,000 \times 0.20 = \text{Rs. } 200$
- Selling Price = $1,000 - 200 = \text{Rs. } 800$

You pay Rs. 800 instead of Rs. 1,000.

Module 18: Annuities and Investments

Author: ZB ZAIB (Module 18)

An annuity is a series of equal payments made at regular intervals. Understanding annuities is crucial for financial planning, loans, and retirement savings.

Types of Annuities:

1. **Ordinary Annuity:** Payments made at the end of each period
2. **Annuity Due:** Payments made at the beginning of each period

Daily Life Example 1: Monthly Rent

You pay Rs. 15,000 rent every month for an apartment. This is an ordinary annuity because you make regular payments at the end of each month.

Daily Life Example 2: Insurance Premiums

You pay Rs. 5,000 every month for life insurance. Over 10 years, you make 120 payments. The insurance company uses annuity calculations to determine your coverage and benefits.

Daily Life Example 3: Car Loan

You take a car loan of Rs. 500,000 at 8% interest. You pay Rs. 10,000 every month for 5 years. This is an annuity payment plan where the bank calculates the total interest you will pay.

Accumulated Value

The accumulated value is the total amount of money you will have at the end of all annuity payments, including interest earned.

Module 19: Introduction to Matrices

Author: ZB ZAIB (Module 19)

A matrix is a rectangular arrangement of numbers in rows and columns. Matrices are powerful tools for organizing and manipulating data.

Definition:

A matrix is denoted by a capital letter and written in brackets. For example:

Plain Text

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

This is a 2×3 matrix (2 rows, 3 columns).

Daily Life Example 1: Student Grades

A teacher has 3 students and 4 subjects. The grades can be organized as a matrix:

Student	Math	English	Science	History
Student1	85	90	88	92
Student2	78	85	90	88
Student3	92	88	85	90

Daily Life Example 2: Store Inventory

A store has 3 branches and 4 types of products. The inventory is organized as a matrix:

Branch	Product1	Product2	Product3	Product4
Branch1	100	150	200	120
Branch2	120	180	190	140
Branch3	110	160	210	130

Module 20: Matrix Operations

Author: ZB ZAIB (Module 20)

Matrix operations include addition, subtraction, and multiplication. These operations help combine or compare different sets of data.

Matrix Addition

Two matrices can be added if they have the same order (same number of rows and columns). Add corresponding elements:

Plain Text

```
[1  2]: # "[5  6]  [6  8]"
[3  4]: # "+ [7  8] = [10 12]"
```

Daily Life Example: Combining Sales Data

Store A sales in Q1: [100] [200] [150]Store B sales in Q1: [120] [180] [160]Total sales in Q1: [220] [380] [310]

Scalar Multiplication

Multiply every element of the matrix by a number:

Plain Text

$$2 \times \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

Daily Life Example: Scaling Production

If a factory produces [100] [200] [150] units of three products, and you want to double production:

$$2 \times [100] [200] [150] = [200] [400] [300]$$

Module 21: Ratios and Proportions

Author: ZB ZAIB (Module 21)

A ratio compares two quantities. A proportion states that two ratios are equal.

Ratio Definition:

A ratio of a to b is written as a:b or a/b.

Daily Life Example 1: Class Composition

A class has 20 boys and 30 girls. The ratio of boys to girls is 20:30, which simplifies to 2:3.

Daily Life Example 2: Recipe Scaling

A recipe for 4 people calls for 2 cups of flour. How much flour is needed for 10 people?

Using proportion:

Plain Text

$$\begin{aligned} 2/4 &= x/10 \\ \text{Cross-multiply: } 2 \times 10 &= 4 \times x \\ 20 &= 4x \\ x &= 5 \text{ cups} \end{aligned}$$

Daily Life Example 3: Currency Conversion

If 1 USD = 280 PKR, how much is 50 USD?

Plain Text

$$1/280 = 50/x$$

$$x = 50 \times 280 = 14,000 \text{ PKR}$$

Module 22: Introduction to Statistics

Author: ZB ZAIB (Module 22)

Statistics is the science of collecting, analyzing, and interpreting numerical data to make decisions and draw conclusions.

Importance of Statistics:

Statistics helps us understand patterns, make predictions, and make informed decisions based on data.

Daily Life Example 1: Weather Forecasting

Meteorologists collect temperature, humidity, and pressure data to predict weather. They use statistical analysis to forecast rain or sunshine.

Daily Life Example 2: Election Polls

Before elections, polling agencies conduct surveys and use statistics to predict which candidate might win.

Daily Life Example 3: Medical Research

Doctors collect patient data and use statistics to determine the effectiveness of a new medicine.

Population vs. Sample:

- **Population:** The entire group of interest
- **Sample:** A smaller subset of the population

Daily Life Example: Quality Control

A factory produces 10,000 light bulbs daily. Testing all 10,000 is expensive, so they test a sample of 100 bulbs to ensure quality.

Module 23: Data Representation

Author: ZB ZAIB (Module 23)

Data representation involves organizing and displaying data visually using charts, graphs, and tables to make it easier to understand.

Types of Data:

- 1. **Qualitative Data:** Describes qualities (color, gender, opinion)
- 2. **Quantitative Data:** Numerical values (height, weight, age)

Methods of Data Representation:

1. Frequency Table (Tabulation)

Organizing data in rows and columns:

Age Group	Frequency
10-20	15
20-30	25
30-40	20
40-50	10

2. Bar Chart

Using rectangular bars to represent data.

Daily Life Example: A chart showing sales of different products (shirts, pants, jackets) with each product represented by a bar.

3. Pie Chart

Showing proportions as slices of a circle.

Daily Life Example: A pie chart showing how a family's monthly income is divided: 40% rent, 30% food, 20% utilities, 10% savings.

4. Line Graph

Showing trends over time.

Daily Life Example: A line graph showing how temperature changes throughout the day, from morning to evening.

5. Histogram

Similar to a bar chart but for continuous data.

Daily Life Example: A histogram showing the distribution of student heights in a class.

Module 24: Central Tendency - Mode

Author: ZB ZAIB (Module 24)

Central tendency measures describe the center or typical value of a data set. The mode is the most frequently occurring value.

Mode Definition:

The value that appears most often in a data set.

Daily Life Example 1: Shoe Sizes in a Store

Shoe sizes sold: 6, 7, 7, 8, 7, 9, 8, 7, 10

The mode is 7 because it appears 4 times, more than any other size.

Daily Life Example 2: Favorite Colors in a Class

Colors chosen: Red, Blue, Red, Green, Red, Blue, Red, Yellow

The mode is Red because it appears 4 times.

Daily Life Example 3: Most Common Salary

In a company, salaries are: 50,000, 60,000, 50,000, 70,000, 50,000, 80,000

The mode is 50,000 because it appears 3 times.

Bimodal and Multimodal:

If two values appear with the same highest frequency, the data is bimodal. If more than two values have the same highest frequency, it is multimodal.

Module 25: Median and Quartiles

Author: ZB ZAIB (Module 25)

The median is the middle value when data is arranged in order. Quartiles divide the data into four equal parts.

Median Definition:

The middle value in an ordered data set.

Daily Life Example 1: House Prices

House prices in a neighborhood: 1M, 2M, 3M, 4M, 5M

The median price is 3M (the middle value).

Daily Life Example 2: Student Heights

Heights of 7 students (in cm): 150, 155, 160, 165, 170, 175, 180

The median height is 165 cm (the middle value).

Quartiles:

- **Q1 (First Quartile):** 25th percentile (25% of data below this value)

- **Q2 (Second Quartile):** 50th percentile (median)
- **Q3 (Third Quartile):** 75th percentile (75% of data below this value)

Interquartile Range (IQR):

Plain Text

$$\text{IQR} = Q3 - Q1$$

Module 26: Geometric and Harmonic Means

Author: ZB ZAIB (Module 26)

These are special types of averages used in specific situations involving growth rates and speeds.

Geometric Mean:

Used for growth rates and situations involving percentages or ratios.

Formula:

Plain Text

$$\text{GM} = \sqrt[n]{(x_1 \times x_2 \times \dots \times x_n)}$$

Daily Life Example 1: Investment Growth

An investment grows 10% in year 1, 20% in year 2, and 15% in year 3. The average growth rate is:

Plain Text

$$\text{GM} = \sqrt[3]{(1.10 \times 1.20 \times 1.15)} = \sqrt[3]{1.518} \approx 1.149 \text{ or } 14.9\% \text{ average growth}$$

Harmonic Mean:

Used for average speeds, rates, and ratios.

Formula:

Plain Text

$$\text{HM} = n / (1/x_1 + 1/x_2 + \dots + 1/x_n)$$

Daily Life Example 1: Average Speed

You travel 100 km at 50 km/h and then 100 km at 100 km/h. The average speed is:

Plain Text

$$HM = 2 / (1/50 + 1/100) = 2 / 0.03 \approx 66.67 \text{ km/h}$$

Note: The arithmetic mean would be 75 km/h, which is incorrect for this scenario.

Module 27: Measures of Dispersion

Author: ZB ZAIB (Module 27)

Dispersion measures how spread out data is from the center. A low dispersion means data is clustered; high dispersion means data is spread out.

Range:

The difference between the highest and lowest values.

Formula:

Plain Text

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Daily Life Example 1: Temperature Variation

Daily temperatures: 15°C, 18°C, 22°C, 25°C, 28°C

$$\text{Range} = 28 - 15 = 13^\circ\text{C}$$

Quartile Deviation:

Half of the interquartile range.

Formula:

Plain Text

$$\text{Quartile Deviation} = (Q3 - Q1) / 2$$

Daily Life Example: Exam Score Spread

$$Q1 = 50, Q3 = 80$$

$$\text{Quartile Deviation} = (80 - 50) / 2 = 15$$

This means the middle 50% of scores are spread over a range of 30 points.

Module 28: Standard Deviation and Variance

Author: ZB ZAIB (Module 28)

Standard deviation measures how much data deviates from the mean. It is one of the most important measures of dispersion.

Variance:

The average of the squared deviations from the mean.

Formula:

Plain Text

$$\text{Variance} = \frac{\sum (x - \text{mean})^2}{n}$$

Standard Deviation:

The square root of the variance.

Formula:

Plain Text

$$\text{SD} = \sqrt{\text{Variance}} = \sqrt{\frac{\sum (x - \text{mean})^2}{n}}$$

Daily Life Example 1: Quality Control

A factory fills juice bottles with a target of 500 ml.

Bottle volumes: 498, 500, 502, 499, 501 ml

Mean = 500 ml

Variance = 2, SD $\approx$ 1.41 ml

This low standard deviation indicates consistent filling.

Coefficient of Variation:

Plain Text

$$\text{CV} = \left(\frac{\text{SD}}{\text{Mean}} \right) \times 100$$

Used to compare variability between different data sets.

Module 29: Permutations and Combinations

Author: ZB ZAIB (Module 29)

Permutations and combinations are counting methods used to find the number of ways to arrange or select items.

Permutation:

An arrangement where the order matters.

Formula:

Plain Text

$${}^n P_r = n! / (n-r)!$$

Daily Life Example 1: Arranging Books

How many ways can you arrange 3 books on a shelf?

Plain Text

$${}^3 P_3 = 3! = 6 \text{ ways}$$

Daily Life Example 2: Selecting Officers

A club has 5 members. How many ways can you select a President, Secretary, and Treasurer?

Plain Text

$${}^5 P_3 = 5! / 2! = 60 \text{ ways}$$

Combination:

A selection where the order does not matter.

Formula:

Plain Text

$${}^n C_r = n! / [r!(n-r)!]$$

Daily Life Example 1: Selecting a Committee

How many ways can you select 3 people from a group of 5 for a committee?

Plain Text

$${}^5 C_3 = 10 \text{ ways}$$

Factorial:

Plain Text

$$n! = n \times (n-1) \times (n-2) \times \dots \times 1$$

Examples: $3! = 6$, $5! = 120$, $0! = 1$

Module 30: Definitions of Probability

Author: ZB ZAIB (Module 30)

Probability is the measure of the likelihood that an event will occur. It ranges from 0 (impossible) to 1 (certain).

Classical Definition:

Plain Text

$$P(\text{Event}) = \text{Number of Favorable Outcomes} / \text{Total Number of Possible Outcomes}$$

Daily Life Example 1: Coin Toss

Probability of getting heads = $1/2 = 0.5$ or 50%

Daily Life Example 2: Dice Roll

Probability of rolling a 3 = $1/6 \approx 16.7\%$

Daily Life Example 3: Drawing a Card

Probability of drawing a red card from a deck = $26/52 = 50\%$

Relative Frequency Definition:

Probability based on observed data from repeated experiments

Daily Life Example: Weather Prediction

If it rained on 30 days out of 100 days in a season, the probability of rain on any given day is 30%.

Mutually Exclusive Events:

Events that cannot occur at the same time.

Daily Life Example 1: Coin Toss

Getting heads and getting tails are mutually exclusive.

Exhaustive Events:

Events that together cover all possible outcomes.

Daily Life Example: Die Roll

The events {1, 2, 3, 4, 5, 6} are exhaustive because they cover all possible outcomes.

Key Conceptual Definitions {#key-conceptual-definitions}

Author: ZB ZAIB (Definitions)

Term	Definition	Daily Life Example
Percentage Change	The relative change between an old value and a new value, expressed as a percentage.	Calculating the increase in your monthly electricity bill from Rs. 5,000 to Rs. 6,000 is a 20% increase.
Simple Interest	Interest paid only on the original principal amount, not on accumulated interest.	Borrowing Rs. 10,000 from a friend at 10% simple interest for 2 years means paying Rs. 2,000 in interest.
Compound Interest	Interest calculated on both the principal and the accumulated interest.	Growth of money in a long-term fixed deposit account where interest is reinvested.
Annuity	A series of equal payments made at regular intervals over a specified period.	Monthly insurance premium payments or monthly rent payments.
Matrix	A rectangular arrangement of numbers in rows and columns.	A cinema seating plan organized by rows and seat numbers.
Ratio	A comparison of two quantities by division.	Mixing 1 part syrup with 4 parts water to make a drink (1:4 ratio).
Proportion	A statement that two ratios are equal.	If 2 cups of flour makes 12 cookies, then 4 cups makes 24 cookies (2:12 = 4:24).
Statistics	The study of collection, analysis, and interpretation of numerical data.	Analyzing weather patterns to predict rain or analyzing election polls.
Qualitative Data	Data that describes qualities or characteristics, not	Color, gender, opinion, or preference.

	numbers.	
Quantitative Data	Data that can be measured and expressed numerically.	Height, weight, age, test scores, or temperature.
Mode	The value that occurs most frequently in a data set.	The most common shoe size sold in a store.
Median	The middle value in a sorted list of numbers.	The middle house number in a street of 11 houses.
Mean	The average of all values in a data set.	The average score of a class on a test.
Range	The difference between the maximum and minimum values.	The difference between the highest and lowest temperature in a day.
Quartile	Values that divide a data set into four equal parts.	Q1 represents the 25th percentile, Q2 the 50th percentile (median), Q3 the 75th percentile.
Geometric Mean	The nth root of the product of n values, used for growth rates.	Calculating the average annual growth rate of an investment over multiple years.
Harmonic Mean	The reciprocal of the average of the reciprocals, used for average speeds.	Calculating the average speed for a journey with different speeds over equal distances.
Standard Deviation	A measure of how much data deviates from the mean on average.	Checking the consistency of the weight of flour bags in a bakery.
Variance	The square of the standard deviation.	If standard deviation is 5, variance is 25.
Coefficient of Variation	The ratio of standard deviation to mean, expressed as a percentage.	Comparing variability between different data sets with different units.
Permutation	An arrangement of items where the order is important.	Setting a 4-digit PIN code for your phone (1234 is different from 4321).
Combination	A selection of items where the order does not matter.	Selecting 11 players for a cricket team from a squad of 15.

Probability	The measure of the likelihood that an event will occur.	The chance of winning a lottery or rolling a specific number on a die.
Mutually Exclusive Events	Events that cannot occur at the same time.	Getting heads or tails in a coin toss (cannot get both simultaneously).
Exhaustive Events	Events that together cover all possible outcomes.	All possible outcomes when rolling a die: {1, 2, 3, 4, 5, 6}.
Factorial	The product of all positive integers up to a number.	$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

100 Quiz Questions with Answers {#100-quiz-questions}

Author: ZB ZAIB (Quiz)

Module 16-17 Questions (1-15)

Q1: What is the formula for calculating percentage change?

- **A:** $(\text{Change} / \text{Original Value}) \times 100$
- **E:** Percentage change measures how much a value has increased or decreased relative to its starting point.

Q2: If a shop's sales increased from Rs. 50,000 to Rs. 75,000, what is the percentage increase?

- **A:** 50%
- **E:** $\text{Change} = 75,000 - 50,000 = 25,000$. $\text{Percentage increase} = (25,000 / 50,000) \times 100 = 50\%$.

Q3: What is simple interest?

- **A:** Interest calculated only on the principal amount, not on accumulated interest.
- **E:** Simple interest is straightforward—you earn interest only on the money you initially invested.

Q4: What is compound interest?

- **A:** Interest calculated on the principal plus any previously earned interest.
- **E:** With compound interest, you earn interest on your interest, leading to exponential growth.

Q5: A bank offers 5% simple interest on a Rs. 10,000 deposit for 3 years. How much interest will you earn?

- **A:** Rs. 1,500
- **E:** Simple Interest = Principal $\times$ Rate $\times$ Time = $10,000 \times 5\% \times 3 = \text{Rs. } 1,500$.

Q6: What is a discount?

- **A:** A reduction in the original price of an item.
- **E:** When a store has a "50% off" sale, they are offering a discount.

Q7: If an item is marked at Rs. 500 and has a 20% discount, what is the selling price?

- **A:** Rs. 400
- **E:** Discount = $500 \times 20\% = \text{Rs. } 100$. Selling Price = $500 - 100 = \text{Rs. } 400$.

Q8: What is the difference between simple and compound interest?

- **A:** Simple interest is calculated only on the principal; compound interest includes interest on accumulated interest.
- **E:** Simple interest grows linearly, while compound interest grows exponentially over time.

Q9: A loan of Rs. 50,000 is taken at 8% simple interest per year. How much will be owed after 2 years?

- **A:** Rs. 58,000
- **E:** Simple Interest = $50,000 \times 8\% \times 2 = \text{Rs. } 8,000$. Total Amount = $50,000 + 8,000 = \text{Rs. } 58,000$.

Q10: If a product's price decreases from Rs. 1,000 to Rs. 750, what is the percentage decrease?

- **A:** 25%
- **E:** Change = $1,000 - 750 = \text{Rs. } 250$. Percentage decrease = $(250/1,000) \times 100 = 25\%$.

Q11: What is an annuity?

- **A:** A series of equal payments made at regular intervals over a specified period.
- **E:** Monthly rent payments, car installments, and insurance premiums are all examples of annuities.

Q12: Why is compound interest more favorable than simple interest for savings?

- **A:** Because you earn interest on your interest, leading to exponential growth.
- **E:** Over time, compound interest generates significantly more wealth than simple interest.

Q13: A store offers a "Buy 2, Get 1 Free" promotion. If each item costs Rs. 100, what is the effective discount percentage?

- **A:** Approximately 33.33%
- **E:** You pay for 2 items (Rs. 200) but get 3 items. The effective price per item is Rs. 66.67, which is a discount of about 33.33%.

Q14: If you invest Rs. 5,000 at 6% annual compound interest, how much will you have after 2 years?

- **A:** Rs. 5,618
- **E:** Year 1: $5,000 \times 1.06 = \text{Rs. } 5,300$. Year 2: $5,300 \times 1.06 = \text{Rs. } 5,618$.

Q15: What is the accumulated value in an annuity?

- **A:** The total value of all payments plus the interest earned on them.
- **E:** If you save Rs. 1,000 every month for a year at 5% interest, the accumulated value includes all 12 payments plus the interest earned on each payment.

Module 18 Questions (16-25)

Q16: What is the future value of an annuity?

- **A:** The total amount of money you will have at the end of making all annuity payments.
- **E:** If you save Rs. 500 monthly for 5 years at 4% annual interest, the future value is the sum of all your contributions plus the interest earned.

Q17: What is the present value of an annuity?

- **A:** The current worth of all future annuity payments, discounted to today's value.
- **E:** If someone promises to pay you Rs. 1,000 annually for 5 years, the present value is less than Rs. 5,000 because money received in the future is worth less today.

Q18: How often are annuity payments typically made?

- **A:** Yearly, semi-annually, quarterly, or monthly.
- **E:** Common payment frequencies are annual, semi-annual, quarterly, and monthly.

Q19: What is an ordinary annuity?

- **A:** An annuity where payments are made at the end of each period.
- **E:** Most loans and savings plans are ordinary annuities. You make your first payment at the end of the first month.

Q20: What is an annuity due?

- **A:** An annuity where payments are made at the beginning of each period.

- **E:** Rent is often paid at the beginning of the month, making it an annuity due.

Q21: If you deposit Rs. 2,000 at the end of each year for 4 years at 5% interest, what is the future value?

- **A:** Approximately Rs. 8,620
- **E:** This is an ordinary annuity. Using the future value formula: $FV = 2,000 \times \left[\frac{(1.05^4 - 1)}{0.05} \right] \approx \text{Rs. } 8,620$.

Q22: What is the discount factor in annuities?

- **A:** A factor used to convert future payments into their present value.
- **E:** The discount factor accounts for the time value of money—money received later is worth less than money received today.

Q23: Why are annuities important for retirement planning?

- **A:** They provide a steady stream of income over a fixed period.
- **E:** Annuities ensure that retirees receive regular payments, helping them plan their finances without worrying about running out of money.

Q24: What is the accumulation factor in an annuity?

- **A:** A factor that shows how much your money will grow over time with interest.
- **E:** The accumulation factor helps calculate how much your regular deposits will grow.

Q25: If an annuity payment is Rs. 1,500 per month for 3 years at 4% annual interest, is this an ordinary annuity or annuity due?

- **A:** It depends on whether payments are made at the beginning or end of each month.
- **E:** If payments are made at the end of each month, it's an ordinary annuity. If at the beginning, it's an annuity due.

Module 19-20 Questions (26-40)

Q26: What is a matrix?

- **A:** A rectangular arrangement of numbers in rows and columns.
- **E:** A matrix is like a table. For example, a 2×3 matrix has 2 rows and 3 columns.

Q27: What is the order of a matrix?

- **A:** The number of rows by the number of columns ($m \times n$).
- **E:** A matrix with 3 rows and 4 columns has an order of 3×4 .

Q28: Can you add two matrices of different orders?

- **A:** No, matrices must have the same order to be added.

- **E:** You can only add a 2×3 matrix to another 2×3 matrix.

Q29: What is the result of adding matrices A and B if $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$?

- **A:** $\begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$
- **E:** Add corresponding elements: $(1+5=6, 2+6=8, 3+7=10, 4+8=12)$.

Q30: What is scalar multiplication of a matrix?

- **A:** Multiplying every element of the matrix by a single number (scalar).
- **E:** If you multiply the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ by 2, you get $\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$.

Q31: What is the identity matrix?

- **A:** A square matrix with 1s on the diagonal and 0s elsewhere.
- **E:** The 2×2 identity matrix is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Q32: Can you multiply a 2×3 matrix by a 3×2 matrix?

- **A:** Yes, the result will be a 2×2 matrix.
- **E:** Matrix multiplication is possible when the number of columns in the first matrix equals the number of rows in the second matrix.

Q33: What is the transpose of a matrix?

- **A:** A new matrix formed by swapping rows and columns.
- **E:** The transpose of $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ is $\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$.

Q34: What is a square matrix?

- **A:** A matrix with the same number of rows and columns.
- **E:** A 3×3 matrix or a 4×4 matrix are square matrices.

Q35: If matrix A is 3×4 and matrix B is 4×2 , what is the order of the product AB?

- **A:** 3×2
- **E:** When multiplying matrices, the resulting matrix has dimensions (rows of A) $\times$ (columns of B) = 3×2 .

Q36: What is the purpose of using matrices in business?

- **A:** To organize and analyze large amounts of data efficiently.
- **E:** A company can use matrices to store sales data for multiple products across different regions.

Q37: What is a zero matrix?

- **A:** A matrix where all elements are zero.

- **E:** $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ is a 2×2 zero matrix.

Q38: Can you subtract matrices of different orders?

- **A:** No, they must have the same order.
- **E:** Just like addition, subtraction requires matrices to have identical dimensions.

Q39: If you multiply a 2×2 matrix by a 2×2 matrix, what is the order of the result?

- **A:** 2×2
- **E:** The result of multiplying two 2×2 matrices is always a 2×2 matrix.

Q40: What is the determinant of a matrix used for?

- **A:** To determine if a matrix is invertible and to solve systems of linear equations.
- **E:** A matrix with a determinant of zero is singular (non-invertible).

Module 21 Questions (41-50)

Q41: What is a ratio?

- **A:** A comparison of two quantities by division.
- **E:** If a class has 20 boys and 30 girls, the ratio of boys to girls is 2:3.

Q42: What is a proportion?

- **A:** A statement that two ratios are equal.
- **E:** If $2:3 = 4:6$, this is a proportion because both ratios are equal.

Q43: If the ratio of apples to oranges is 3:2 and there are 15 apples, how many oranges are there?

- **A:** 10 oranges
- **E:** If 3 apples correspond to 2 oranges, then 15 apples correspond to 10 oranges.

Q44: What is a direct proportion?

- **A:** A relationship where as one quantity increases, the other increases proportionally.
- **E:** If you buy 2 kg of sugar for Rs. 200, then 4 kg costs Rs. 400.

Q45: What is an inverse proportion?

- **A:** A relationship where as one quantity increases, the other decreases proportionally.
- **E:** If 10 workers can complete a job in 5 days, then 5 workers will take 10 days.

Q46: If a recipe calls for 2 cups of flour for 12 cookies, how much flour is needed for 36 cookies?

- **A:** 6 cups of flour

- **E:** Using proportion: $2/12 = x/36$. Solving: $x = 6$ cups.

Q47: What is the cross-multiplication method in proportions?

- **A:** A method where you multiply the numerator of one ratio by the denominator of the other.
- **E:** For the proportion $a/b = c/d$, cross-multiplication gives $a \times d = b \times c$.

Q48: If a car travels 150 km in 3 hours, how far will it travel in 5 hours at the same speed?

- **A:** 250 km
- **E:** Speed = $150/3 = 50$ km/h. Distance in 5 hours = $50 \times 5 = 250$ km.

Q49: What is the ratio of 45 minutes to 2 hours?

- **A:** 3:8
- **E:** Convert 2 hours to 120 minutes. The ratio is 45:120, which simplifies to 3:8.

Q50: If the ratio of boys to girls in a school is 4:5 and there are 360 students total, how many girls are there?

- **A:** 200 girls
- **E:** Total parts = $4 + 5 = 9$. Girls = $(5/9) \times 360 = 200$.

Module 22-24 Questions (51-65)

Q51: What is statistics?

- **A:** The science of collecting, analyzing, and interpreting numerical data.
- **E:** A cricket analyst uses statistics to determine which batsman is performing best.

Q52: What is the difference between population and sample in statistics?

- **A:** A population is the entire group; a sample is a subset of the population.
- **E:** If you want to know the average height of all students in Pakistan, the population is all Pakistani students.

Q53: What is the mode?

- **A:** The value that appears most frequently in a data set.
- **E:** In the data set {2, 3, 3, 5, 7, 3, 9}, the mode is 3 because it appears three times.

Q54: What is the mean?

- **A:** The average of all values in a data set.
- **E:** The mean of {2, 4, 6, 8} is $(2+4+6+8)/4 = 5$.

Q55: What is the median?

- **A:** The middle value when data is arranged in order.
- **E:** In the data set {1, 3, 5, 7, 9}, the median is 5 (the middle value).

Q56: Can a data set have more than one mode?

- **A:** Yes, if multiple values appear with the same highest frequency.
- **E:** In {1, 1, 2, 2, 3, 4}, both 1 and 2 are modes (bimodal).

Q57: What is the range of a data set?

- **A:** The difference between the highest and lowest values.
- **E:** In {5, 10, 15, 20, 25}, the range is $25 - 5 = 20$.

Q58: What is a frequency distribution?

- **A:** A table showing how often each value or group of values occurs.
- **E:** A frequency distribution might show that 10 students scored 80-90, 15 scored 70-80.

Q59: What is a histogram?

- **A:** A bar chart used to display the distribution of continuous data.
- **E:** A histogram showing student test scores might have bars for score ranges like 60-70, 70-80, 80-90.

Q60: What is a pie chart used for?

- **A:** To show the proportion of different categories as parts of a whole.
- **E:** A pie chart might show that 40% of a company's budget goes to salaries, 30% to operations.

Q61: What is qualitative data?

- **A:** Data that describes qualities or characteristics, not numbers.
- **E:** Color, gender, and opinion are qualitative data.

Q62: What is quantitative data?

- **A:** Data that can be measured and expressed numerically.
- **E:** Height, weight, age, and test scores are quantitative data.

Q63: What is the purpose of data representation?

- **A:** To make data easier to understand and interpret visually.
- **E:** A graph showing sales trends over time is easier to understand than a table of numbers.

Q64: What is a bar chart?

- **A:** A chart using rectangular bars to represent data values.

- **E:** A bar chart might show the sales of different products.

Q65: What is a line graph used for?

- **A:** To show trends or changes in data over time.
- **E:** A line graph showing temperature changes throughout the day helps visualize how temperature varies.

Module 25-26 Questions (66-75)

Q66: What is the median in a frequency distribution?

- **A:** The value that divides the distribution into two equal parts.
- **E:** In a frequency distribution of test scores, the median score is the one where 50% of students scored below and 50% scored above.

Q67: What are quartiles?

- **A:** Values that divide a data set into four equal parts.
- **E:** Q1 (first quartile) is at 25%, Q2 (median) is at 50%, and Q3 (third quartile) is at 75%.

Q68: What is the interquartile range (IQR)?

- **A:** The difference between the third and first quartiles ($Q3 - Q1$).
- **E:** If $Q1 = 40$ and $Q3 = 60$, then $IQR = 60 - 40 = 20$.

Q69: What is the geometric mean?

- **A:** The n th root of the product of n values.
- **E:** The geometric mean of 2, 4, and 8 is $\sqrt[3]{(2 \times 4 \times 8)} = \sqrt[3]{64} = 4$.

Q70: When is the geometric mean used?

- **A:** For growth rates and situations involving percentages or ratios.
- **E:** To calculate the average growth rate of an investment over several years.

Q71: What is the harmonic mean?

- **A:** The reciprocal of the average of the reciprocals of the values.
- **E:** The harmonic mean is used to calculate average speeds.

Q72: When is the harmonic mean used?

- **A:** For average speeds, rates, and ratios.
- **E:** To find the average speed for a journey with different speeds over equal distances.

Q73: What is the relationship between arithmetic mean, geometric mean, and harmonic mean?

- **A:** Arithmetic Mean $\geq$ Geometric Mean $\geq$ Harmonic Mean (for positive values).
- **E:** For any positive numbers, the arithmetic mean is always greater than or equal to the geometric mean.

Q74: What is a percentile?

- **A:** A value below which a certain percentage of observations fall.
- **E:** The 90th percentile means 90% of the data falls below this value.

Q75: What is the empirical relation between mean, median, and mode?

- **A:** Mode $\approx 3(\text{Median}) - 2(\text{Mean})$ for moderately skewed distributions.
- **E:** This relationship helps estimate the mode if you know the mean and median.

Module 27-28 Questions (76-85)

Q76: What is dispersion?

- **A:** The spread or variability of data around a central value.
- **E:** Two classes might have the same average score of 50, but one class has scores ranging from 45-55 (low dispersion).

Q77: What is the range?

- **A:** The difference between the maximum and minimum values.
- **E:** In $\{10, 20, 30, 40, 50\}$, the range is $50 - 10 = 40$.

Q78: What is the quartile deviation?

- **A:** Half of the interquartile range (IQR/2).
- **E:** If IQR = 20, then quartile deviation = $20/2 = 10$.

Q79: What is the mean deviation?

- **A:** The average of the absolute deviations from the mean.
- **E:** For $\{2, 4, 6, 8\}$, mean = 5. Mean deviation = $(3+1+1+3)/4 = 2$.

Q80: What is standard deviation?

- **A:** A measure of how much data deviates from the mean on average.
- **E:** A low standard deviation means data is clustered near the mean.

Q81: What is variance?

- **A:** The square of the standard deviation.
- **E:** If standard deviation = 5, then variance = 25.

Q82: What is the coefficient of variation?

- **A:** The ratio of standard deviation to mean, expressed as a percentage.
- **E:** $CV = (\text{Standard Deviation} / \text{Mean}) \times 100$.

Q83: Why is standard deviation more useful than range?

- **A:** Because it considers all values, not just the extremes.
- **E:** Range only looks at the highest and lowest values.

Q84: What does a coefficient of variation of 20% mean?

- **A:** The standard deviation is 20% of the mean.
- **E:** This indicates moderate variability in the data.

Q85: How is standard deviation calculated?

- **A:** Take the square root of the variance (average of squared deviations from the mean).
- **E:** $SD = \sqrt{[\Sigma(x - \text{mean})^2 / n]}$.

Module 29-30 Questions (86-100)

Q86: What is a permutation?

- **A:** An arrangement of items where the order matters.
- **E:** Arranging 3 people (A, B, C) in a line: ABC, ACB, BAC, BCA, CAB, CBA are 6 different permutations.

Q87: What is a combination?

- **A:** A selection of items where the order does not matter.
- **E:** Selecting 2 people from {A, B, C}: {A,B}, {A,C}, {B,C} are 3 combinations.

Q88: What is the formula for permutations?

- **A:** $nPr = n! / (n-r)!$
- **E:** The number of permutations of 5 items taken 3 at a time: $5P3 = 60$.

Q89: What is the formula for combinations?

- **A:** $nCr = n! / [r!(n-r)!]$
- **E:** The number of combinations of 5 items taken 3 at a time: $5C3 = 10$.

Q90: What is a factorial?

- **A:** The product of all positive integers up to a number ($n! = n \times (n-1) \times \dots \times 1$).
- **E:** $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

Q91: What is probability?

- **A:** The measure of the likelihood that an event will occur (0 to 1).

- **E:** The probability of rolling a 6 on a die is $1/6 \approx 0.167$ or 16.7%.

Q92: What are mutually exclusive events?

- **A:** Events that cannot occur at the same time.
- **E:** In a coin toss, getting heads and getting tails are mutually exclusive.

Q93: What are exhaustive events?

- **A:** Events that together cover all possible outcomes.
- **E:** In a die roll, the events $\{1, 2, 3, 4, 5, 6\}$ are exhaustive.

Q94: What is the classical definition of probability?

- **A:** $P(\text{Event}) = \text{Number of favorable outcomes} / \text{Total number of possible outcomes}$.
- **E:** $P(\text{rolling a 3}) = 1/6$.

Q95: What is the relative frequency definition of probability?

- **A:** Probability based on observed data from repeated experiments.
- **E:** If a coin lands on heads 52 times out of 100 tosses, the relative frequency probability is 0.52.

Q96: How many ways can you arrange 4 books on a shelf?

- **A:** 24 ways
- **E:** $4! = 4 \times 3 \times 2 \times 1 = 24$.

Q97: How many ways can you choose 3 students from a class of 10 for a committee?

- **A:** 120 ways
- **E:** ${}^{10}C_3 = 10! / (3! \times 7!) = 120$.

Q98: What is the probability of drawing a red card from a standard deck of 52 cards?

- **A:** $26/52 = 1/2 = 0.5$ or 50%
- **E:** There are 26 red cards (13 hearts + 13 diamonds) out of 52 total cards.

Q99: What is the difference between theoretical and experimental probability?

- **A:** Theoretical is based on mathematical calculation; experimental is based on actual results.
- **E:** Theoretical probability of heads in a coin toss is 0.5. Experimental might be 0.48 if you toss 100 times.

Q100: If you roll two dice, what is the probability of getting a sum of 7?

- **A:** $6/36 = 1/6 \approx 0.167$ or 16.7%

- **E:** The favorable outcomes are (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) = 6 outcomes out of 36 possible.
-

Study Tips and Recommendations {#study-tips}

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Effective Study Strategies

1. Understand Before Memorizing

Do not memorize formulas without understanding their meaning. For each formula, learn why it works and when to apply it. This understanding will help you solve new problems even if you forget the exact formula.

2. Practice Daily Life Examples

Connect mathematical concepts to real-world situations. The daily life examples provided in this document are designed to help you see the practical application of each concept. Try to think of your own examples as well.

3. Work Through Problems Step by Step

When solving problems, show all your work step by step. This helps you identify where you might have made a mistake and allows you to learn from errors.

4. Review Regularly

Mathematics requires regular review and practice. Spend time each day reviewing previously learned concepts and solving practice problems.

5. Form Study Groups

Studying with classmates can help you understand concepts better. Explain concepts to others, and ask them to explain concepts to you. Teaching is one of the best ways to learn.

6. Use Visual Aids

Create diagrams, charts, and tables to visualize concepts. Visual representations often make abstract concepts easier to understand.

Preparation for Exams

1. Start Early

Begin your exam preparation at least two weeks before the exam. This gives you enough time to cover all material and practice problems.

2. Create a Study Schedule

Divide the material into manageable sections and create a schedule for studying each section. Stick to your schedule to ensure you cover all material.

3. Practice with Sample Questions

Use the 100 quiz questions provided in this document to practice. Try to solve them without looking at the answers first, then check your work.

4. Focus on Weak Areas

Identify the concepts or topics where you struggle the most and focus extra time on these areas.

5. Get Adequate Sleep

Do not stay up late studying the night before the exam. Adequate sleep is crucial for memory retention and mental clarity.

6. Review on Exam Day

On the day of the exam, review key formulas and concepts one more time. Avoid learning new material on exam day.

During the Exam

1. Read Questions Carefully

Read each question carefully to understand what is being asked. Sometimes, misunderstanding the question leads to incorrect answers.

2. Manage Your Time

Allocate time for each question based on its difficulty and the total time available. Do not spend too much time on one question.

3. Show Your Work

Always show your calculations and reasoning. Even if your final answer is wrong, you may receive partial credit for showing correct steps.

4. Check Your Answers

If time permits, review your answers and check your calculations.

Conclusion

This comprehensive preparation document provides all the essential material needed to master MTH001 modules 16 to 30. By studying the explanations, understanding the definitions, and practicing the quiz questions, you will develop a strong foundation in these mathematical and statistical concepts.

Remember, mathematics is not just about memorizing formulas—it is about understanding concepts and applying them to solve real-world problems. The daily life examples provided throughout this document are designed to help you see the practical relevance of each concept.

Success in mathematics requires:

- Regular practice and review
- Understanding concepts deeply
- Connecting theory to real-world applications
- Seeking help when needed
- Maintaining a positive attitude toward learning

We wish you the best in your studies and exams. May your hard work and dedication lead to success.

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*"Insaan ke liye wahi kuch hota hai jis ke liye woh khud mehnat aur koshish karta hai."***"For a person, only that which they strive and work hard for comes to pass."*— Surah An-Najm, Ayah 39